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Worcester Polytechnic Institute

Presentation Template

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SECTION 01

Motivation

Why does this problem matter?

01

Background

A brief summary of prior work and relevant context.

- ▶ **Key point one** – establish the setting
- ▶ **Key point two** – supporting evidence or physical intuition
- ▶ **Key point three** – what this work builds on

Narrative text flows naturally here. You can include inline math, e.g. the shear distortion tensor $\gamma = \gamma_1 + i\gamma_2$, without a dedicated math slide. Here is a citation: [1]

Method Overview

Architecture

- ▶ Input: galaxy stamp I , PSF stamp P
- ▶ Two convolutional branches
- ▶ Fused via cross-attention layer
- ▶ Output: $\hat{\gamma}_1, \hat{\gamma}_2$

Loss

$$\mathcal{L} = \sum_i w_i (\hat{\gamma}_i - \gamma_i)^2$$

Results



Figure 1: Placeholder – replace with your plot or diagram.

Variational Loss

We minimise the Tikhonov-regularised functional:

$$\mathcal{L}[f] = \int_{\Omega} |\nabla f|^2 \, d\Omega + \lambda \|f - g\|^2$$

where $\lambda > 0$ controls regularisation and g is the observed signal.

“

Say something profound.

— Alice Gaehring

SECTION 02

Analysis & Discussion

02

Discussion

Summary of main findings and their interpretation.

1. First conclusion: tied directly to the data
2. Second conclusion: broader implication
3. Open questions that remain

Takeaway: A highlighted callout box: useful for the single key message you want the audience to leave with.

Acknowledgements

- | | |
|-----------------------|---|
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Thank you for your time!

Questions?

- [1] S. C. Brenner and L. R. Scott, "Basic Concepts," in *The Mathematical Theory of Finite Element Methods*, New York, NY: Springer New York, 2008, pp. 1–22. doi: 10.1007/978-0-387-75934-0_1.